Model Results Review: New Temporal Models Compared with the Report

Report: Data Driven Prediction of Local Government Water Service Delivery Outcomes through Machine Learning (N Hlumbene)
Sources reviewed: notebook *01_data_loading_and_merging.ipynb* (old model), *train_financial_risk.py* and *retrain_financial_risk_rolling.py* with their result files (new models), the *thesis_figures* folder, and *Final Report (Chapter 3 revised).docx*.
Date: 1 November 2026

Short answer. You are right. The new results do not support a claim that the model predicts accurately, and they do not support any claim about predicting *water service delivery outcomes*. Once the model is tested honestly (trained on earlier years, tested on later years, using only the previous year's data), the Random Forest:

- is **less accurate than simply labelling every municipality "poor"** (82,9% against 85,2% in the rolling backtest);
- correctly recognises only **about half of the "good" municipalities** (specificity 0,56), or fewer than a third when tuned to catch nearly all "poor" ones;
- is only **slightly better than carrying forward last year's label** (balanced accuracy 0,72 against 0,71);
- does show **real but moderate ranking ability** (ROC AUC 0,82), so there is some signal in the financial data.

The report's headline figures (91,7% accuracy, F1 0,940, ROC AUC 0,967) cannot be reproduced from the notebook, came from a design with leakage, and should be withdrawn.

1. What the models actually predict

Every model, old and new, predicts the same label. A municipality year is labelled **"poor" (1)** if irregular expenditure exceeds R10 million, unauthorised expenditure exceeds R10 million, or fruitless and wasteful expenditure exceeds R1 million. Otherwise it is labelled "good" (0). This is a **financial governance risk label**. No water service variable (interruptions, access, Blue Drop scores) is used to define it, and no test has shown that the label tracks water service delivery. The project's own figure notes state: "All current models predict the project's threshold based financial governance risk label. They do not predict water service outcomes."

2. How the new models differ from the old one

Aspect	Old model (notebook, used in the report)	New models (temporal scripts)
Features	Same year financial data, including irregular expenditure trend features that contain the target year's values, plus the municipality code and the year	Previous year (t minus 1) financial data only, with no municipality code or year
Train and test split	Random 70/30 split of rows, so the same municipality appears in both sets in neighbouring years	Time ordered: (a) train on 2013 to 2020, test on 2021 to 2024; (b) rolling backtest, where each year from 2019 to 2024 is predicted by a model trained only on earlier years
Rows with no expenditure data	509 rows with no irregular, unauthorised or fruitless expenditure data were labelled "good". They make up 46% of all "good" labels	Excluded, because their true label is unknown

Missing feature values	Filled with zeros. KNN imputation was imported but never applied	Median imputation inside the model pipeline, plus missing value indicators
Tuning	Fixed settings, with random 5 fold cross validation	Grid search with year ordered (expanding window) validation
Decision threshold	Fixed at 0,5	0,5 in the temporal holdout; in the rolling backtest, either a balanced accuracy threshold or a high recall threshold (aiming for at least 95% of poor cases caught), chosen on training years only
Comparison baselines	None	Majority baseline (call everything "poor") and prior year persistence (repeat last year's label)
Metrics	Accuracy, precision, recall, F1, ROC AUC	Adds balanced accuracy, specificity for "good" cases, PR AUC and confusion matrices

3. Results of the new models

3.1 Temporal holdout (train 2013 to 2020, test 2021 to 2024; 852 cases, 84,5% "poor")

Model	Accuracy	Balanced accuracy	Recall (poor)	Specificity (good)	F1 (poor)	ROC AUC	PR AUC
Majority baseline (all "poor")	0,845	0,500	1,000	0,000	0,916	n/a	n/a
Prior year persistence	0,856	**0,720**	0,917	**0,523**	0,915	n/a	n/a
Logistic Regression	0,845	0,500	1,000	0,000	0,916	0,728	0,940
Decision Tree	0,863	0,588	0,986	0,189	0,924	0,781	0,935
Random Forest	0,863	0,600	0,981	0,220	0,923	**0,826**	0,959

What this means: at the default 0,5 threshold, Logistic Regression predicts "poor" for every single municipality, so it is identical to the majority baseline. The Random Forest is only 1,8 percentage points more accurate than that baseline, and it correctly identifies just 29 of the 132 "good" cases (22%). The simple rule "repeat last year's label" has a much higher balanced accuracy (0,72 against 0,60). The ROC AUC of 0,83 shows the Random Forest's probabilities do rank risk better than chance, but the 0,5 cutoff turns this into almost blanket "poor" predictions.

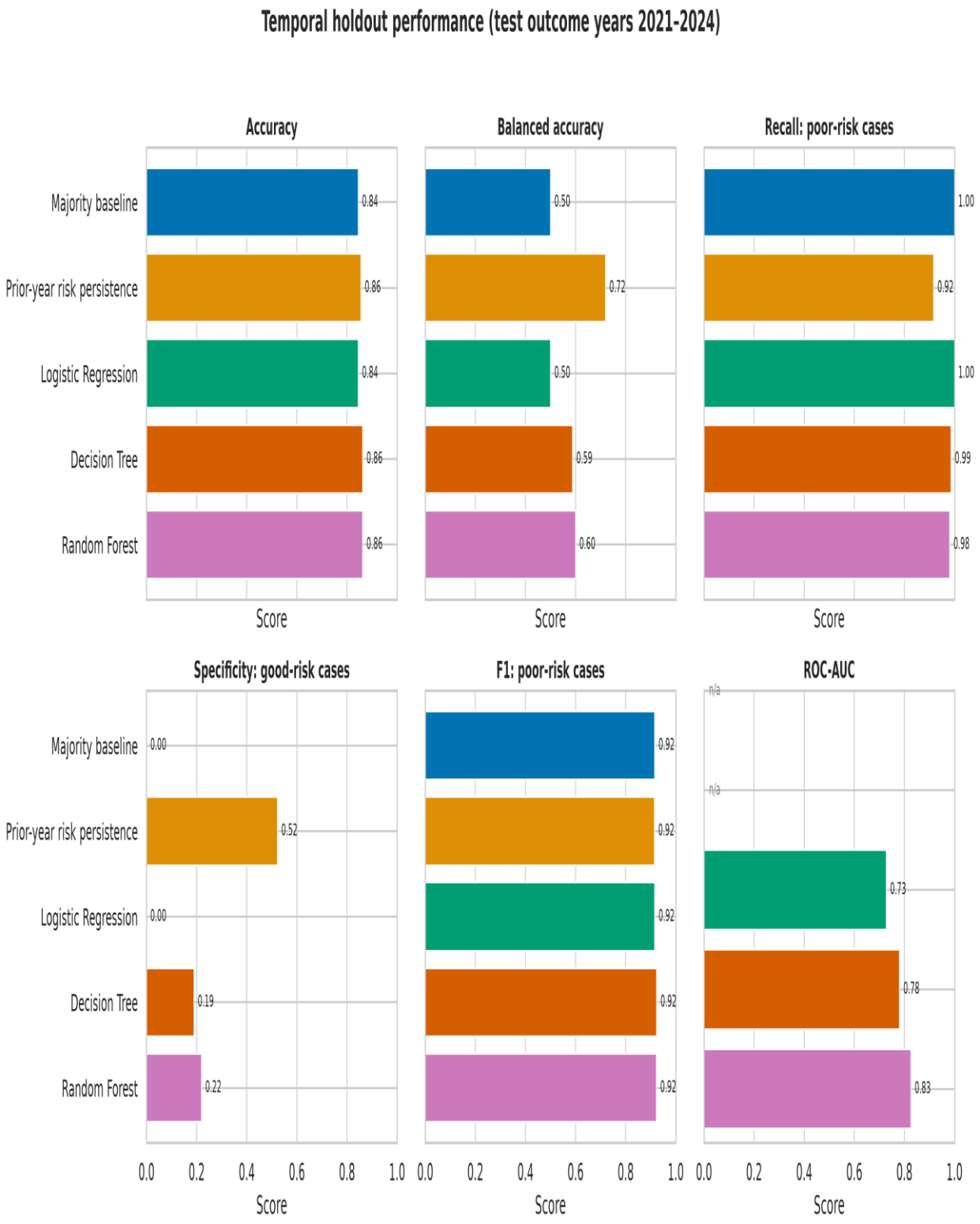


Figure A (thesis_figures/fig02). Temporal holdout scores. Accuracy and F1 look high and almost identical for every model, including the baseline that predicts "poor" for everyone. The differences only show in balanced accuracy and specificity, where prior year persistence performs best.

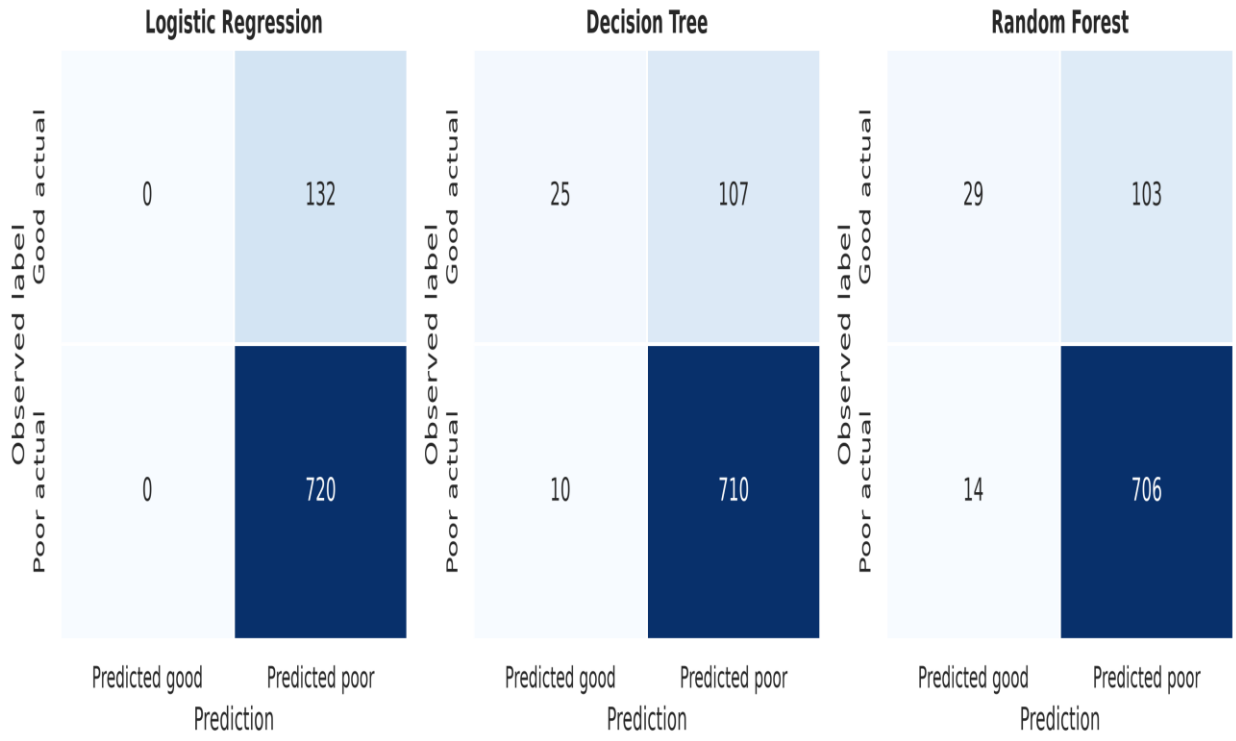


Figure B (thesis_figures/fig07). Temporal holdout confusion matrices at the 0,5 cutoff. Logistic Regression: 0 good cases recognised, 132 false alarms. Decision Tree: 25 recognised, 107 false alarms. Random Forest: 29 recognised, 103 false alarms, 14 missed poor cases.

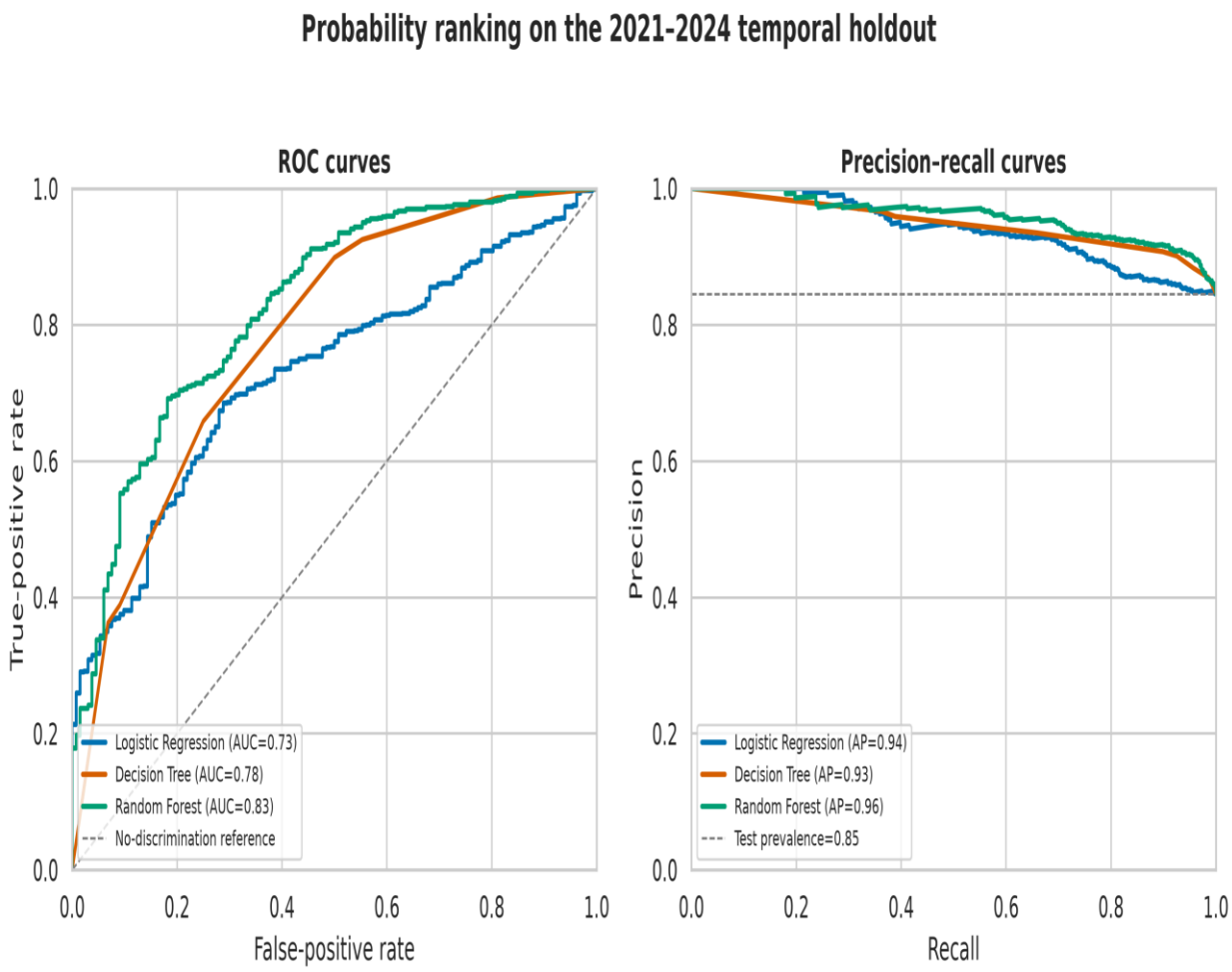


Figure C (thesis_figures/fig08). ROC and precision recall curves for the temporal holdout. The Random Forest curve (ROC AUC 0,83) sits clearly above the diagonal, so the model has some ranking ability. In the precision recall panel, every curve starts from a 0,85 floor (the share of "poor" cases), so a PR AUC of 0,96 is a smaller improvement than it looks.

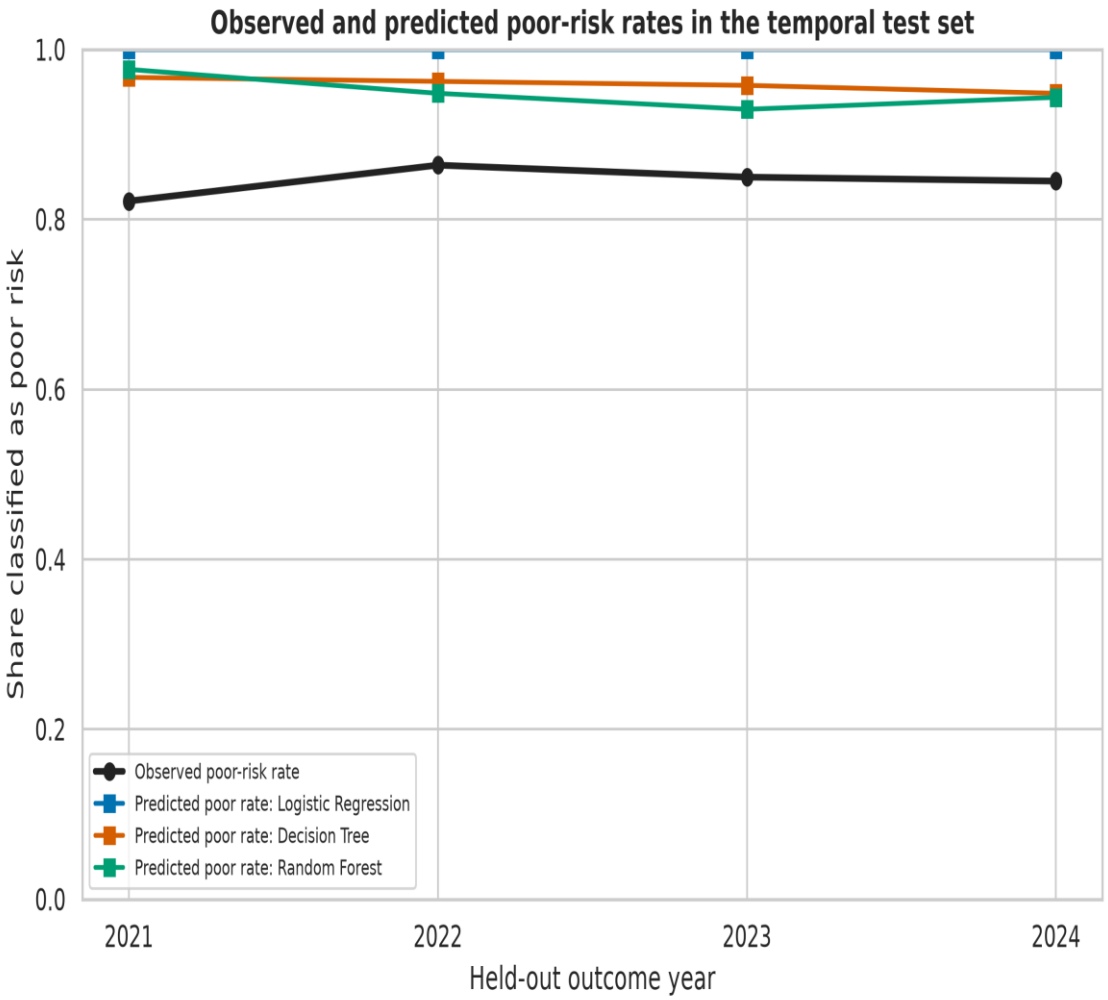


Figure D (thesis_figures/fig10). Observed against predicted share of "poor" by test year. Observed rates were 82% to 86%. The Random Forest predicted 93% to 98% "poor" and Logistic Regression predicted 100% every year, so the models systematically over predict the majority class.

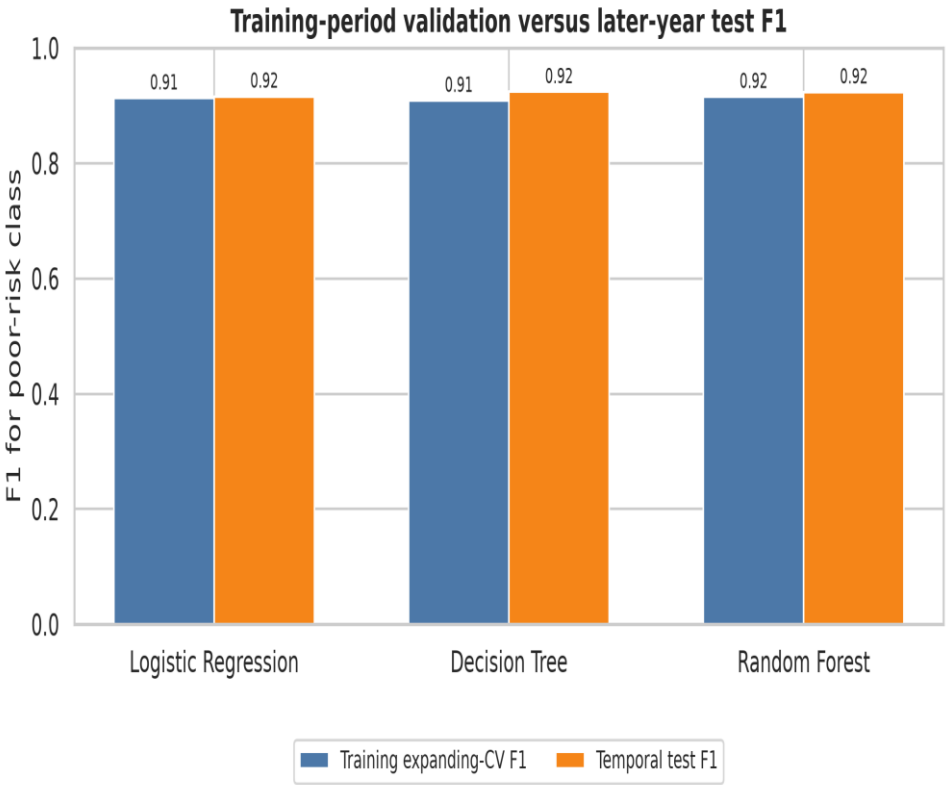


Figure E (thesis_figures/fig09). Validation F1 during training (about 0,91) and test F1 (about 0,92) are almost the same. This consistency is not evidence of a good model: the majority baseline also scores 0,916, so F1 cannot separate useful models from useless ones on this data.

3.2 Rolling backtest (each year from 2019 to 2024 predicted from earlier years; 1 278 cases, 85,2% "poor")

Model and threshold	Accuracy	Balanced accuracy	Recall (poor)	Specificity (good)	ROC AUC	Missed poor / false alarms
Majority baseline	**0,852**	0,500	1,000	0,000	n/a	0 / 189
Prior year persistence	0,850	0,706	0,910	0,503	n/a	98 / 94
Logistic Regression, balanced	0,767	0,627	0,826	0,429	0,736	190 / 108
Decision Tree, balanced	0,799	0,696	0,842	0,550	0,740	172 / 85
Random Forest, balanced	0,829	**0,718**	0,875	**0,561**	**0,820**	136 / 83
Logistic Regression, high recall	0,828	0,529	0,953	0,106	0,736	51 / 169
Decision Tree, high recall	0,847	0,585	0,958	0,212	0,740	46 / 149
Random Forest, high recall	0,868	0,634	0,966	0,302	0,820	37 / 132

What this means: the Random Forest is consistently the best of the three algorithms, and it is the only one whose balanced accuracy beats prior year persistence, but only by about one percentage point (0,718 against 0,706). With the balanced threshold, its overall accuracy (82,9%) is below the majority baseline (85,2%). With the high recall threshold it catches 96,6% of poor cases, but only by flagging 132 of the 189 good cases (70%) as poor as well. You cannot get both high recall and good specificity from these data: improving one costs the other.

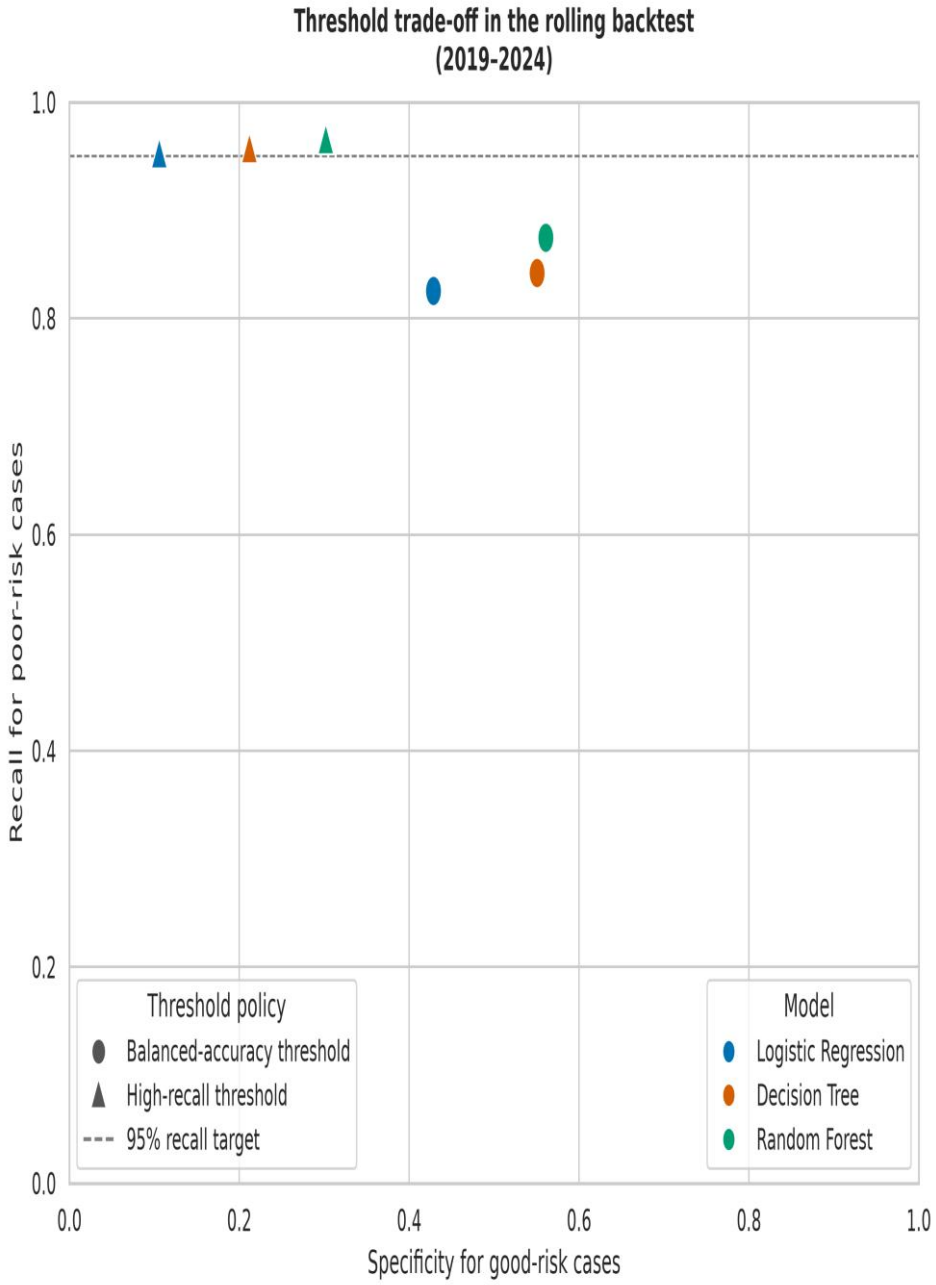


Figure F (thesis_figures/fig03). The trade off between catching poor cases (recall) and recognising good cases (specificity). No model reaches the top right corner, which is where an accurate classifier would sit.

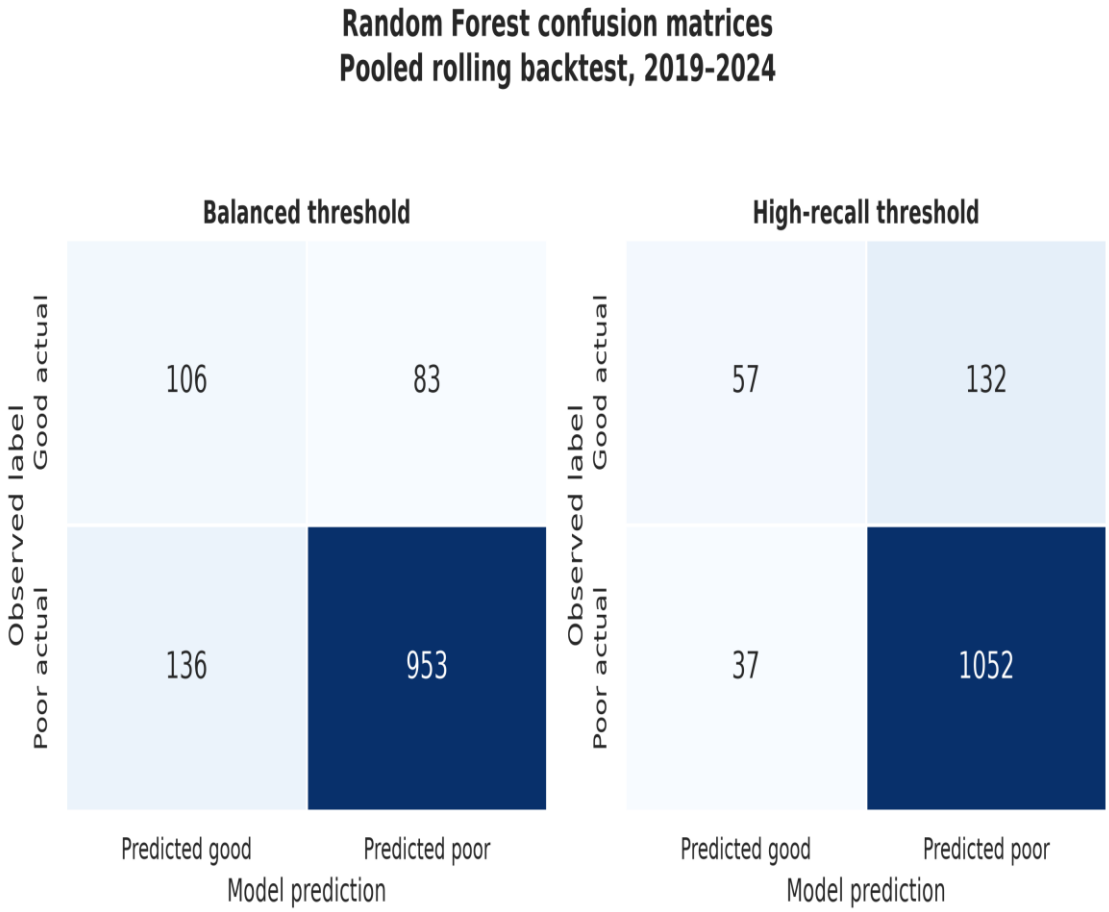


Figure G (thesis_figures/fig05). Random Forest pooled confusion matrices from 2019 to 2024. Balanced threshold: 106 of 189 good cases recognised, 136 poor cases missed. High recall threshold: only 57 good cases recognised, 37 poor cases missed.

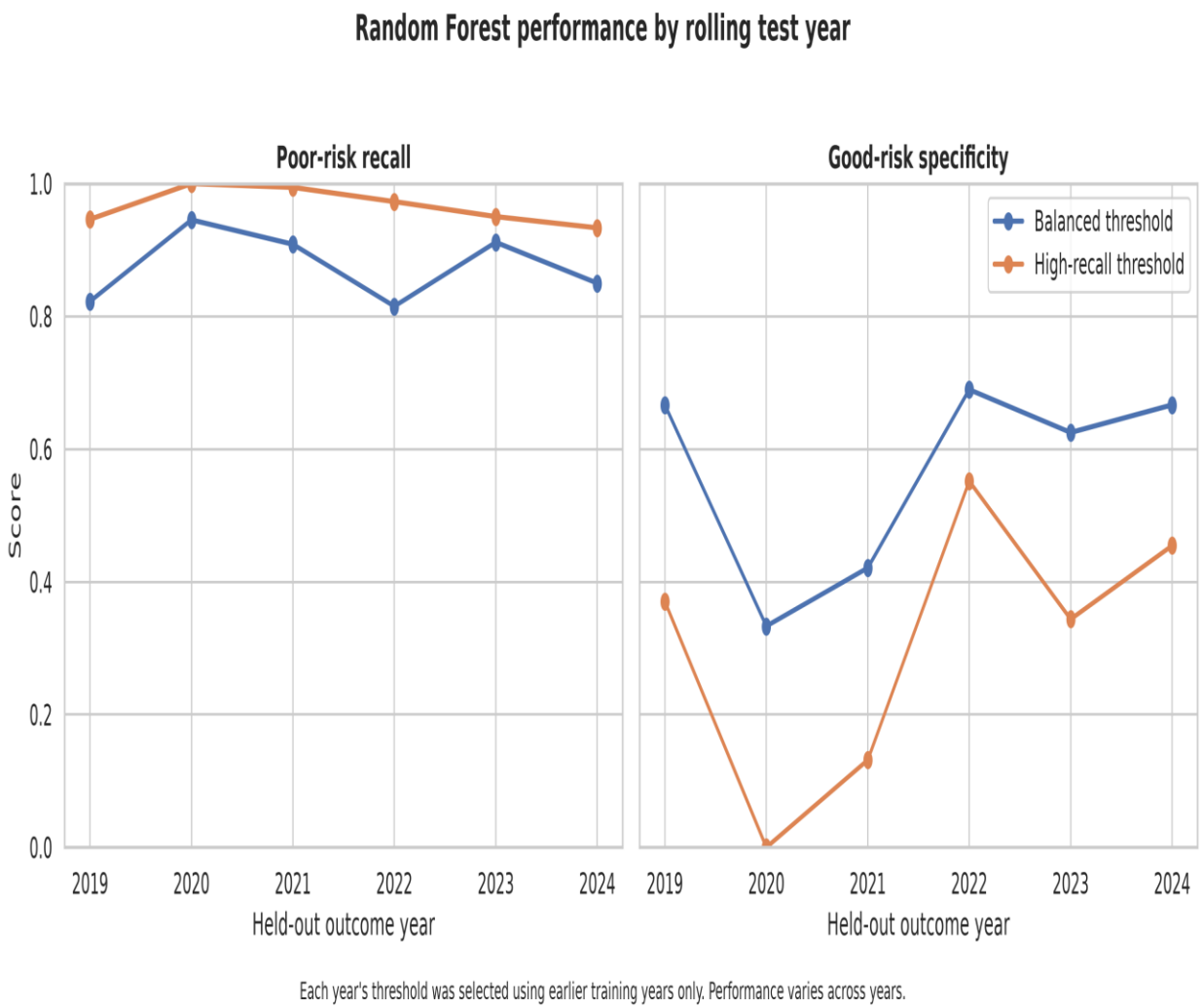


Figure H (thesis_figures/fig04). Random Forest performance by test year. Specificity swings from 0,33 (2020) to 0,69 (2022) with the balanced threshold, and falls to 0,00 in 2020 with the high recall threshold, when the model predicted every municipality as poor. Performance is unstable from year to year, which is typical of small samples.

4. Why the old results looked so much better

#	Cause	Explanation
1	**Reported figures are not reproducible**	The report's numbers (accuracy 0,917, F1 0,940, ROC AUC 0,967) are typed by hand into a summary table in notebook cell 25. The notebook's recorded runs produced different results. The first run had leakage and scored accuracy 0,994 and ROC AUC 1,000. The run after removing the leaking columns, which is the one the report describes, scored accuracy 0,872, F1 0,906 and ROC AUC 0,942. No saved run produces the reported figures.
2	Residual leakage	After the three raw expenditure columns were removed, the features *irreg_exp_change* and *irreg_exp_3yr_avg* were kept. Both are calculated from the same year's irregular expenditure, which is part of the label definition.
3	Memorising municipalities	The random split put the same municipality in training and testing in neighbouring years, and the municipality code and year were used as features. The model could learn "this municipality is usually poor" rather than any general warning sign. This is why SHAP ranked municipality identity as the strongest predictor.

4	Missing data labelled "good"	509 rows with no expenditure data were labelled good, with their features filled with zeros. The model could learn "missing data means good", which is an artefact of the data and not a real pattern.
5	Misleading metrics	With roughly 70% to 85% "poor" labels, accuracy and F1 stay high even for a model that predicts "poor" for everyone. Chicco and Jurman (2020:1) and Saito and Rehmsmeier (2015:1), both now cited in Chapter 3, warn about exactly this.

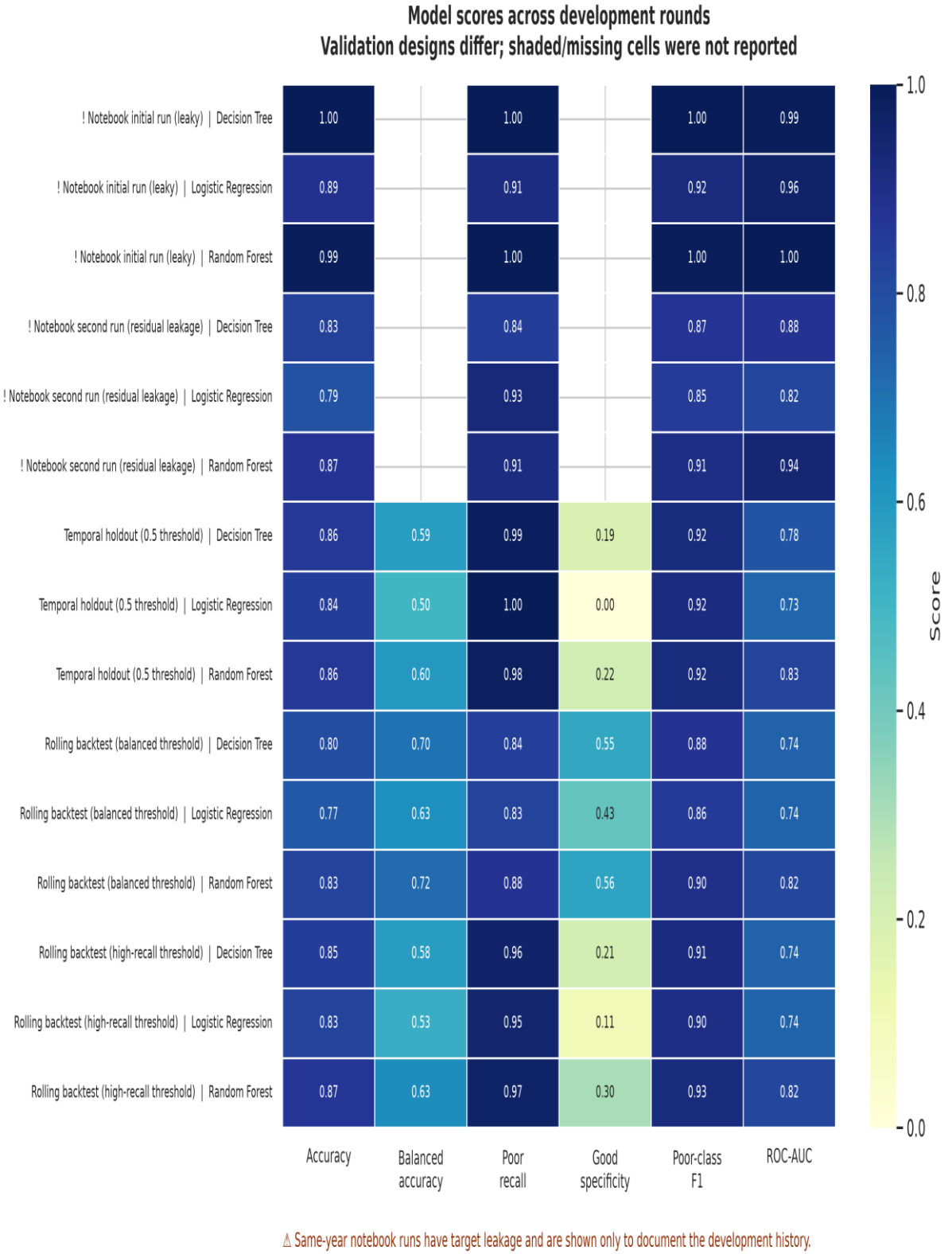


Figure I (thesis_figures/fig01). Scores across every development round. As the leakage is removed and the testing becomes more realistic, accuracy drops from 0,99 to about 0,83 and ROC AUC from 1,00 to about 0,82. Balanced accuracy and specificity, which the old runs never reported, are much lower.

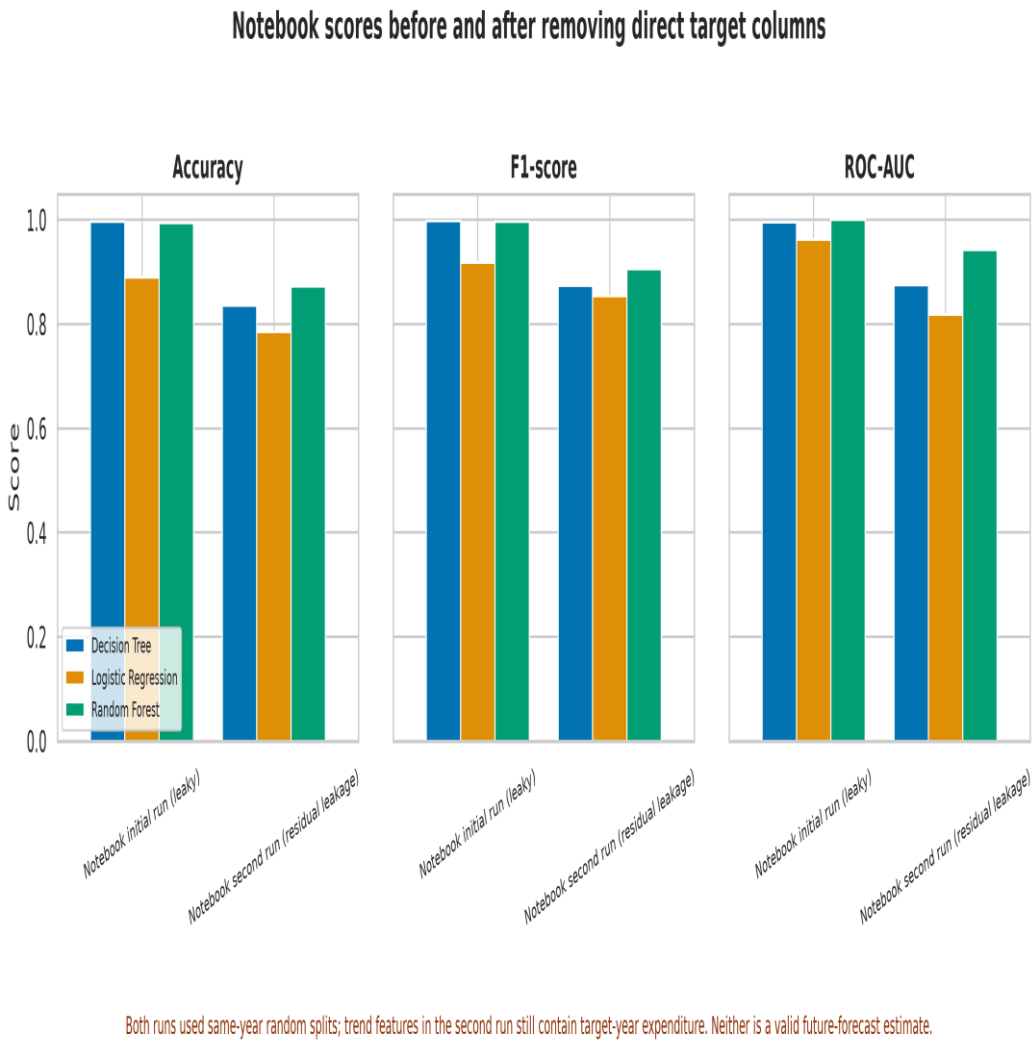


Figure J (thesis_figures/fig06). Scores from the old notebook's two runs. Both used same year random splits, so they should be presented only as development history, not as evidence of predictive ability.

5. The limited data problem in numbers

Measure	Value
Rows in the merged dataset	3 375 (292 municipalities, 2012 to 2025)
Rows with an unknown label (no expenditure data)	509
Rows usable for honest next year prediction	2 617 (known label and previous year data)
Municipalities usable per year from 2016	213 of 292
"Good" cases per test year from 2019 to 2024	27 to 38 (189 in total)
Share of "poor" labels in usable data	81% overall; 82% to 87% from 2019 onwards

Missing values in previous year features	Water revenue 73%, waste water revenue 72%, debt impairment 77%, repairs and maintenance 64%, capital expenditure 55%
Direct water service outcome data	None (no interruption, access or Blue Drop data linked to the label)

The models have very few "good" examples to learn from, and the water related financial variables are missing for most municipalities. This explains the low specificity, the year to year instability and the small gain over simple baselines. Prior year persistence performs almost as well because a municipality's financial governance status changes little from one year to the next.

6. Claims in the report compared with the evidence

Claim in the report	Where	What the evidence shows	Status
Random Forest achieved accuracy 91,7%, F1 0,940 and ROC AUC 0,967	Abstract; 4.5.1 (Table 2); 4.7; 5.2.1 RQ2; 5.2.2 table	Not reproducible from the notebook; came from a leaky design. Under honest testing: accuracy 0,829 to 0,868, balanced accuracy 0,60 to 0,72, ROC AUC 0,82 to 0,83	**Not supported: replace**
"Consistent five fold cross validation results confirming the reliability of these findings"	Abstract	Random row cross validation has the same leakage. Year ordered validation F1 (0,915) equals what the majority baseline achieves	**Not supported**
Recall of 96,3% is the "success rate on identifying actual poor water service delivery outcomes"	4.5.1	High recall comes from predicting most cases as poor; specificity is 0,22 to 0,56. The label measures financial governance, not water service	**Misleading**
"This study proved that machine learning algorithms can be used to predict water service delivery outcomes with good accuracy"; "accurately predict"	4.7; 5.2.1 Main RQ; 5.4 Conclusion	The target is a financial label never validated against water outcomes; performance is only slightly above simple baselines	**Not supported**
SHAP shows municipality identity, multiyear irregular expenditure and financial year as the strongest predictors of poor water service delivery	Abstract; 4.5.3; 4.6.2; 5.2.1 RQ3	From the leaky random split model that used municipality code and year as features. Identity reflects memorisation. SHAP has not been rerun on the new models	**Not supported: rerun SHAP**
KNN imputation with k = 5, and exclusion of variables with more than 20% missing values	Ch1 E OBJ1; 4.3.6; 5.2.1 RQ1; 5.3; also Ch3 3.6 (written to match the report)	KNN was imported but never applied; missing values were set to zero. Features with 55% to 77% missing values were kept. The new models use median imputation with missing indicators	**Not supported: correct the text**
Target operationalised as a composite of water access, supply interruptions and Blue Drop risk	4.2.2	Target is defined only by three expenditure thresholds	**Not supported: rewrite**

Hyperparameters tuned by grid search	4.2.4	Not done in the old model (fixed settings). Done in the new models with year ordered validation	True only for the new models
Internal validity maintained by removing data leakage	4.5.2	Residual leakage through same year trend features, municipality code and random split	**Partly true**
External validity from 292 municipalities, 9 provinces and 14 years	Abstract; 4.5.2	Only 213 municipalities per year (2 617 rows) usable with honest testing	**Overstated**
Model "resolves the accuracy interpretability trade off" and gives "audit ready risk factor breakdowns"; integrates "operational risk metrics"	5.3; 5.4	No operational water data were used; the accuracy is modest	**Not supported**
The target is a financial governance proxy rather than a direct measure of service delivery	Abstract (final paragraph)	Correct, and the most important limitation	**Supported: strengthen**
Machine learning "may offer a meaningful complement" to audit based monitoring by enabling earlier identification	Abstract	Partly: ROC AUC of 0,82 shows real ranking signal one year ahead, but classification gains over persistence are small	**Supported if worded cautiously**

Note on the latest edits: in the previous revision, Chapter 5 was corrected to match Chapter 4 (91,7%, ROC AUC 0,967), and Chapter 3 Section 3.6 describes the 20% threshold and KNN with five neighbours. These now need to change too, because the Chapter 4 figures and data preparation description are themselves unsupported.

7. What the study can legitimately claim

Suggested wording for the abstract and conclusions:
"Using only the previous year's publicly available municipal financial data, a Random Forest classifier ranked municipalities by next year financial governance risk better than chance (ROC AUC 0,82 in a rolling backtest from 2019 to 2024). However, its classification performance was modest. At a balanced threshold it identified 87,5% of poor risk cases but only 56,1% of good risk cases, a balanced accuracy of 0,72 compared with 0,71 for simply repeating the previous year's label, and its overall accuracy (82,9%) was below that of classifying every municipality as poor (85,2%). Because the target is a threshold based financial governance label and the data are sparse and imbalanced, the model cannot be claimed to predict water service delivery outcomes accurately. It is better understood as an exploratory screening approach that shows financial records contain some early warning signal."

This reframing still answers your research questions honestly. RQ2 (comparing the algorithms) has a clear answer: Random Forest performed best on every threshold independent measure. The main research question can be answered as "partly, with important limitations", which is a legitimate and examinable finding.

8. Recommended changes to the report

#	Change

1	Replace Chapter 4 Table 2 and Figure 9 with the temporal holdout and rolling backtest results (Tables 3.1 and 3.2 above), including both baselines, balanced accuracy and specificity. Use figures fig02, fig04, fig05 and fig08 from thesis_figures.
2	Rewrite 4.2.2 to describe the actual target (three expenditure thresholds) and explain why it is used as a proxy for water service delivery risk.
3	Correct 4.3.6 and related text (Ch1 E OBJ1, Ch3 3.6, Ch5 RQ1 and 5.3) to describe the real data preparation: unknown labels excluded, median imputation with missing value indicators, previous year features.
4	Add a short section explaining the move from the random split to temporal validation and why the earlier scores were inflated. This turns the leakage problem into a methodological contribution, consistent with Kapoor and Narayanan (2023:1).
5	Rerun SHAP on the new Random Forest (previous year features, without the municipality code) before reporting any risk factors.
6	Revise the abstract, 4.7, 5.2.1, 5.2.2, 5.3 and 5.4 to use the cautious wording in Section 7, and add the limited data figures from Section 5 to the limitations.
7	Consider changing the title, for example to "Predicting Municipal Financial Governance Risk as an Early Warning Indicator for Water Service Delivery: A Machine Learning Approach", so that it matches what the model actually predicts.
8	Report the 2019 to 2024 rolling backtest as development evidence, not as an untouched final test, because those years were examined while building the models. A newly labelled year (for example 2025) would provide a genuinely new test.
9	Do not present the Blue Drop analysis (fig11 to fig13) as validation of the model. It compares the stored label, not model predictions, and some municipality code mappings are questionable. It can stay as descriptive context only.

Appendix: figures not used above

Figures fig11, fig12 and fig13 in thesis_figures show Blue Drop 2023 scores by province, their distribution and paired 2014 and 2023 scores for 14 Eastern Cape authorities. They describe water quality context but say nothing about how the models performed, so they are not part of this comparison.